Rohit Singh

AI & SOFTWARE ENGINEERING INTERN

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I'm a **CSE (AI & ML) undergraduate** (Graduation: **2029**) who loves turning complex ideas into working software. Instead of just following tutorials, I focus on building complete, dependable systems from scratch — whether that's training deep learning models with explainable Grad-CAM heatmaps, coding algorithmic **A* graph pathfinding** for robotics, detecting zero-day cyber anomalies, or engineering full-stack platforms with Next.js and WebSockets. I take pride in writing clean, well-tested code, understanding the real failure modes of ML models, and building software that genuinely helps developers learn and build faster.

TECHNICAL SKILLS

Languages:	Python, TypeScript, JavaScript, SQL, HTML5/CSS3.
Machine Learning & AI:	TensorFlow, Keras, scikit-learn, CNNs, Grad-CAM Visual Explainability, Isolation Forest, Random Forest, Ridge Regression, Time-Series Modeling, NumPy, Pandas, Matplotlib.
Core CS & Algorithms:	Data Structures & Algorithms (DSA), A* Graph Search, Dynamic 2D Occupancy Grids, Heuristic Search, Raycasting, Object-Oriented Programming (OOP).
Full-Stack & Systems:	React, Next.js 14, Node.js, Express, Flask, REST APIs, WebSockets (Socket.IO), Tailwind CSS.
Databases & Tools:	PostgreSQL, Supabase, MongoDB, Docker, Git, GitHub Actions, VS Code.

FEATURED PROJECTS

AI Autonomous Navigation & Pathfinding System | Python, A* Search, Heuristics, Dynamic Occupancy Grids 2026

- Engineered an autonomous ground robot navigation stack from scratch, implementing **A* graph traversal** with line-of-sight raycast smoothing for real-time collision-free path planning.
- Constructed dynamic **2D occupancy grid mapping** combined with ultrasonic raycasting perception and pure-pursuit trajectory control.
- Tested across **60 benchmark obstacle layouts** with a **100% goal arrival rate and zero collisions**, proving solid algorithmic edge-case handling under strict computational constraints.

PneumoAI: Diagnostic Vision & Explainability | Python, TensorFlow, Keras, Grad-CAM, Flask 2026

- Trained and validated a deep convolutional neural network on chest radiograph datasets to detect pneumonia, achieving **95.4% validation accuracy** and an AUC-ROC of **0.98**.
- Integrated **Grad-CAM (Gradient-weighted Class Activation Mapping)** to produce visual attention heatmaps, providing clinician-grounded explainability rather than treating the model as a black box.
- Deployed as a standalone Flask application with client-side image upload and sub-second inference visualization.

AI Cybersecurity Threat Detection System | Python, scikit-learn, Isolation Forest, Random Forest, Docker 2026

- Built a defense-in-depth network intrusion detection platform pairing supervised **Random Forest classification (99.2% accuracy)** with unsupervised **Isolation Forest** to flag zero-day anomalous network traffic.
- Mapped detected attacks to the **MITRE ATT&CK** framework with structured forensic telemetry, feeding alerts to an interactive dark-themed SOC analyst console.

AI Energy Consumption Forecasting & Optimization | Python, scikit-learn, Ridge, MLP, Time-Series Backtesting 2026

- Built an hourly demand forecasting platform benchmarking Ridge Regression, Random Forest, and Scaled MLPs against seasonal baselines over **52,000+ hours** of building and household energy datasets.
- Ensured strict **zero-data-leakage chronological validation** and implemented an interactive virtual load-shifting simulator under Time-of-Use tariffs with empirical uncertainty bands (±95%).

CabGo: Real-Time Transportation Platform | Next.js 14, TypeScript, Supabase, WebSockets, PostgreSQL • *(In Active Development)* 2026

- Architecting a production-style cab booking platform with distinct Rider, Driver, and Admin workflows using **Next.js 14, TypeScript, and Supabase**.
- Implementing real-time location telemetry over **WebSockets**, Stripe payment intents, and PostgreSQL Row-Level Security (RLS) policies.

KEY ACHIEVEMENTS & MILESTONES

- **IIT Delhi AI & ML Project Portfolio:** Engineered and completed 5 comprehensive AI/ML platforms across robotics pathfinding, medical imaging, network threat detection, and energy forecasting with rigorous validation.
- **Published & Deployed 4+ Live Applications:** All major projects feature publicly accessible live web demos on GitHub Pages, transparent documentation, and reproducible open-source repositories.
- **Algorithmic & Heuristic Problem Solving:** Designed and benchmarked a 100% collision-free A* graph pathfinder across 60 rigorous simulation maps, demonstrating strong DSA and spatial reasoning.
- **Full-Stack Builder:** Independently developed production-grade TypeScript and Python applications with responsive UX, database models, and real-time socket communication.

EDUCATION

Bachelor of Technology in Computer Science & Engineering (AI & ML)

Hyderabad, India

Undergraduate • Expected Graduation: **2029**

Relevant Coursework: Data Structures & Algorithms, Design and Analysis of Algorithms, Artificial Intelligence, Machine Learning, Database Management Systems (DBMS), Object-Oriented Programming (OOP).